Shengyi Qian

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Contact

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Employment

Research Scientist, Meta FAIR. 2024 -
Focus: Multimodal LLM, Agents, Unified Models.

Research Intern, NVIDIA Robotics Lab. 09/2023 - 04/2024
Supervisor: Ankit Goyal, Kaichun Mo, Valts Blukis, Dieter Fox.

Applied Scientist Intern, Amazon Web Services. 05/2023 - 08/2023
Supervisor: Weifeng Chen, Erran Li, Zhuowen Tu.

Research Intern, Meta FAIR. 05/2021 - 12/2021
Supervisor: Georgia Gkioxari.

Research Interests

I work on unified vision-language models that integrate 3D spatial understanding, and on turning those models into agents that can ground language, perceive affordances, and act in the real world — from robotic manipulation to UI control.

Education

University of Michigan. Ann Arbor, MI
Ph.D. in Computer Science and Engineering. 2019 - 2024
- Advisors: David Fouhey, Joyce Chai.
- Dissertation: Learning to Interact with the 3D World.

University of Michigan. Ann Arbor, MI
B.S.E. in Computer Science, *Summa Cum Laude.* 2017 - 2019

Shanghai Jiao Tong University. Shanghai, China
B.S.E. in Electrical and Computer Engineering. 2015 - 2017

Publications

(* indicates equal contribution)

[1] **Hierarchical Procedural Meta-Reasoning for Generalizable Multimodal Agents.**
Yao Fu, Shengyi Qian, Pierluca D'Oro, Fanyi Xiao, Honglak Lee, Joseph Tighe, Manchen Wang.
ICML 2026.

[1] **Beyond Language Modeling: An Exploration of Multimodal Pretraining.**
Shengbang Tong*, David Fan*, John Nguyen*, Ellis Brown, Gaoyue Zhou, Shengyi Qian, Boyang Zheng, Théophane Vallaeys, Junlin Han, Rob Fergus, Naila Murray, Marjan Ghazvininejad, Mike Lewis, Nicolas Ballas, Amir Bar, Michael Rabbat, Jakob Verbeek, Luke Zettlemoyer†, Koustuv Sinha†, Yann LeCun†, Saining Xie†.
ICML 2026.

[2] **Beyond 3D VQAs: Injecting 3D Spatial Priors into Vision-Language Models for Enhanced Geometric Reasoning.**
Chun-Hsiao Yeh, Shengyi Qian, Manchen Wang, Yi Ma, Joseph Tighe, Fanyi Xiao.
CVPR 2026.

[3] **Circuit Tracing in Vision-Language Models: Understanding the Internal Mechanisms of Multimodal Thinking.**
Jingcheng Yang*, Tianhu Xiong*, Shengyi Qian, Klara Nahrstedt, Mingyuan Wu.
CVPR 2026 (Findings).

[4] **DigiData: Training and Evaluating General-Purpose Mobile Control Agents.**
Yuxuan Sun*, Manchen Wang*, Shengyi Qian*, William R. Wong*, Eric Gan*, Pierluca D'Oro*, Alejandro Castillejo Munoz, Sneha Silwal, Pedro Matias, Nitin Kamra, Satwik Kottur, Nick Raines, Xuanyi Zhao, Joy Chen, Joseph Greer, Andrea Madotto, Allen Bolourchi, James Valori, Kevin Carlberg, Karl Ridgeway, Joseph Tighe.
arXiv 2025.

[5] **DISCO Balances the Scales: Adaptive Domain- and Difficulty-Aware Reinforcement Learning on Imbalanced Data.**
Yuhang Zhou*, Jing Zhu*, Shengyi Qian, Zhuokai Zhao, Xiyao Wang, Xiaoyu Liu, Ming Li, Paiheng Xu, Wei Ai, Furong Huang.
EMNLP 2025 (Findings).

[6] **3D-MVP: 3D Multiview Pretraining for Robotic Manipulation.**
Shengyi Qian, Kaichun Mo, Valts Blukis, David Fouhey, Dieter Fox, Ankit Goyal.
CVPR 2025.

[7] **Mosaic of Modalities: A Comprehensive Benchmark for Multimodal Graph Learning.**
Jing Zhu, Yuhang Zhou, Shengyi Qian, Zhongmou He, Tong Zhao, Neil Shah, Danai Koutra.
CVPR 2025.

[8] **3D-GRAND: A Million-Scale Dataset for 3D-LLMs with Better Grounding and Less Hallucination.**
Jianing (Jed) Yang*, Xuweiyi Chen*, Nikhil Madaan, Madhavan Iyengar, Shengyi Qian, David Fouhey, Joyce Y. Chai.
CVPR 2025.

[9] **Multi-Object Hallucination in Vision-Language Models.**
Xuweiyi Chen*, Ziqiao Ma*, Xuejun Zhang*, Sihan Xu, Shengyi Qian, Jianing Yang, David Fouhey, Joyce Chai.
NeurIPS 2024.

[10] **LinkGPT: Teaching Large Language Models To Predict Missing Links.**

Shengyi Qian, Kaichun Mo, Valts Blukis, David Fouhey, Dieter Fox, Ankit Goyal.
CIKM 2025.

[11] **AffordanceLLM: Grounding Affordance from Vision Language Models.**
Shengyi Qian, Weifeng Chen, Xiong Zhou, Min Bai, Zhuowen Tu, Li Erran Li.
OpenSun3D Workshop @ CVPR 2024.

[12] **LLM-Grounder: Open-Vocabulary 3D Visual Grounding with Large Language Model as an Agent.**
Jianing (Jed) Yang*, Xuweiyi Chen*, Shengyi Qian, Nikhil Madaan, Madhavan Iyengar, David Fouhey, Joyce Y. Chai.
ICRA 2024.

[13] **Pitfalls in Link Prediction with Graph Neural Networks: Understanding the Impact of Target-link Inclusion & Better Practices.**
Jing Zhu*, Yuhang Zhou*, Vassilis N. Ioannidis, Shengyi Qian, Wei Ai, Xiang Song, Danai Koutra.
WSDM 2024.

[14] **Understanding 3D Object Interaction from a Single Image.**
Shengyi Qian, David Fouhey.
ICCV 2023.

[15] **Sound Localization from Motion: Jointly Learning Sound Direction and Camera Rotation.**
Ziyang Chen, Shengyi Qian, Andrew Owens.
ICCV 2023.

[16] **Understanding 3D Object Articulation in Internet Videos.**
Shengyi Qian, Linyi Jin, Chris Rockwell, Siyi Chen, David Fouhey.
CVPR 2022.

[17] **Recognizing Scenes from Novel Viewpoints.**
Shengyi Qian, Alexander Kirillov, Nikhila Ravi, Devendra Singh Chaplot, Justin Johnson, David Fouhey, Georgia Gkioxari.
arXiv 2021.

[18] **Planar Surface Reconstruction from Sparse Views.**
Linyi Jin, Shengyi Qian, Andrew Owens, David Fouhey.
ICCV 2021.

[19] **Associative3D: Volumetric Reconstruction from Sparse Views.**
Shengyi Qian*, Linyi Jin*, David Fouhey.
ECCV 2020.

[20] **OASIS: A Large-Scale Dataset for Single-Image 3D in the Wild.**
Weifeng Chen, Shengyi Qian, David Fan, Noriyuki Kojima, Max Hamilton, Jia Deng.

CVPR 2020.

[21] **Learning Single-Image Depth from Videos using Quality Assessment Networks**.
Weifeng Chen, Shengyi Qian, Jia Deng.
CVPR 2019.

Awards

MLoG Best Paper Award, WSDM 2024.	2024
Rackham Doctoral Intern Fellowship.	2023
Rackham Conference Travel Grant.	2023
James B. Angell Scholar.	2019
EECS Scholar Award.	2018, 2019
Summer Undergraduate Research Award.	2019
Junyuan Scholarship.	2017
Undergraduate Excellent Scholarship.	2016
Distinguished Achievement in Extracurricular Activity.	2016

Talks

Learning to Interact with the 3D World.
Guest Lecture, Cornell University, 11/2024.

Teaching Foundation Models to Interact with the 3D World.
Meta FAIR, 05/2024.

Building an Interactive 3D World from Images.
Roblox, 04/2024.

Understanding 3D Object Interaction from Ordinary Images.
LG AI Research, 02/2024.
NVIDIA Research, 02/2024.
NYC Vision Day (5-min lightning talk), 04/2024.

Understanding 3D Object Interaction from a Single Image.
CVPR 2023 Workshop on 3D Vision and Robotics, 06/2023.

Understanding 3D Object Articulation in Internet Videos.
CVPR 2022, 06/2022.
AI New Youth, 07/2022.

Associative3D: Volumetric Reconstruction from Sparse Views.
ECCV 2020, 08/2020.
Holistic Scene Structures for 3D Vision Workshop at ECCV 2020, 08/2020.

Service

Reviewer:
IJCV, RA-L, CVPR 2021-, ICCV 2021-, ECCV 2022-, NeurIPS 2021-,
ICML 2022-, 3DV 2022-, WACV 2023-, ICLR 2024.

Workshop and Tutorial Organization:
Organizer, 3D-LLM/VLA: Bridging Language, Vision and Action in 3D Environments,

CVPR 2026.
Organizer, 3D-LLM/VLA: Bridging Language, Vision and Action in 3D Environments, CVPR 2025.

Workshop Program Committee:
LangRob @ CoRL 2023.

Student Volunteer: CVPR 2019.

Department Service:
University of Michigan Ph.D. Admissions Committee, AY 2021-2024.

Teaching

EECS442 Computer Vision, University of Michigan. 01/2019 - 04/2019
Instructional Aide (IA).
Instructor: David Fouhey.

VE280 Programming & Data Structure, SJTU. 05/2019 - 08/2019
Teaching Assistant (TA).
Instructor: Paul Weng, Weikang Qian.